Yash Niraj Mehta		**22B1504**
Energy Science and Engineering		**B.Tech.**
Indian Institute of Technology Bombay		**Gender: Male**
		DOB: 25/08/2004

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	

Completed a **Minor Degree** in **Data Science,** offered by Centre of Machine Intelligence and Data Science (cMInDS) at IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Selected among CBSE's top 50 meritorious students to attend **Republic Day 2021** from the **PM's Box** for scoring 99.6% in 10th class
- Secured an All India Rank of **2203** in JEE Advanced'22 and a percentile of **99.976** in the MHT-CET-PCM'22 exam among 0.23 million applicants

PROFESSIONAL EXPERIENCE

American Express India Pvt. Ltd.
Business Strategy Intern
[May'25-Jul'25]

*Global payment services leader with **133M+** cards in force, offering credit cards, charge cards and diverse travel services*

- Devised an offer rollout strategy for a free AmEx product MCG, to increase revenue via targeted recommendations
- Built a segmentation framework using **SQL** on MCG usage data and offer specific attributes to target **US customers**
- Found target segments using **Pivot Tables** to suggest placements for 3 offers: CreditSecure, X-Sell & Personal Loans
- Gauged take-up raise by **20–40%** (CreditSecure), **11–22%** (X-Sell) & origination share rise by **2–5%** (Personal Loans)

RESEARCH EXPERIENCE

Project Intern
[May'24-Jul'24]

Battery Energy Storage System (BESS) Techno-Commercial Analysis | Guide: Prof. P. Sunthar, Chemical Engg, IIT Bombay

- Studied key engineering economics (**NPV**, **Payback**, **TCO**) and global BESS adoption to assess feasibility in India
- Developed **5** models to evaluate BESS strategies for **reducing grid** and **DG set reliance** as well as for **peak shifting**
- Derived simplified expressions for **Savings** & **Levelised Cost of Energy** (**LCOE**), both standalone and with solar PV

B. Tech. Project (EN491)
[Aug'25-Present]

Parameter Estimation of Induction Machines | Guide: Prof. Ravi Prakash Reddy Siddavatam, Energy Engineering, IIT Bombay

- Engineering **Kalman Filter** to estimate induction machine parameters using measured input voltage & current

KEY PROJECTS

Data Quality Improvement
Prof. Vinay Kulkarni
[Oct'23]

Improved transformer current quality via noise analysis, cleaning and predictive reconstruction | Course Project (DS203)

- Analyzed noise thresholds for **80K+** entries of **280 days** transformer current to identify missing, good and bad days
- Enhanced 'good' day data by replacing noisy segments (troughs) with **moving averages**, thus refining data quality
- Built a polynomial regression model (**R^2 = 0.834**) using **backward elimination** to reconstruct bad and missing days

Wine Classification using FTIR Analysis
[Dec'23-Jan'24]

*Classified 37 red wine samples using **K-Means** on **FTIR spectra** | WiDS 2023-24, Analytics Club and DAV Team, IIT Bombay*

- Performed data preprocessing (spectral trimming, scaling, thresholding) and applied **PCA**, retaining **99%+** variance
- Optimized clustering using dual-dataset analysis (original and averaged spectra), achieving accurate 2 category split

Chemical Operating Plant Data Analysis
Prof. Vinay Kulkarni
[Oct'23-Nov'23]

Analyzed 240+ plant features to predict vibrations, rank parameters and model energy demand | Course Project (DS203)

- Cleaned data, engineered features & conducted key multicollinearity checks (**VIF > 6**) for robust regression
- Built robust **MLR** models for vibrations (**84–98%** accuracy), enabling a powerful 4-level safety classification

Hydrogen Train
Prof. Pratibha Sharma
[Feb'24-Apr'24]

*Collaborated in a team of **3** to assess the techno-economic feasibility of hydrogen trains in India | Course Project (EN610)*

- Mapped hydrogen refuelling stations along **3 Mumbai–Delhi** routes based on H_2 demand for **fuel cells \ HICEs**
- Proposed a **Rs.150 Cr** setup, a **20MW** solar PV capacity & area of **50 hectares** for each solar-powered HRS

Switch Energy Case
Energy Transition Plan for Guatemala
[Sept'24-Oct'24]

*Curated **$4.8 Bn** mitigation plan for Guatemala by comparing **6 energy poverty factors** with Ecuador | Case Competition*

- Devised a detailed **national ethanol plan** covering cookstove distribution, **refinery setup**, and E10 integration
- Proposed **PAYG solar** with storage and small hydro power to **boost grid resiliency** and **rural electrification** by **28%**
- Assessed the impact of our plan on 6+ SDGs, estimating **30k+ new jobs** and **80%+ cut in hazardous emissions**

Warehouse Model
Prof. Saurabh Jain
[Jan'25-Apr'25]

*Simulated a real warehouse system in **AnyLogic** using **9,500+** data points of IBD and OBD goods | Course Project (IE630)*

- Fitted probability distributions on inter-arrival & service times; validated parameters via statistical tests (ex: **KS Test**)
- Compared separate vs. shared service block models; identified trade-off in **processing time** & **server utilization**

POSITIONS OF RESPONSIBILITY

DAMP Mentor
SMP, IIT Bombay
[Jun'25-Present]

*Selected among **11/39** Department Academic Mentorship Program Applicants based on peer reviews and interviews*

- Mentoring **12** sophomores to help them excel in their academic, career-centric and extra-curricular activities
- Part of a 5-member subgroup responsible for updating and maintaining the DAMP Blog catering to **300+ UGs**

Institute BioX Convenor
ITC, IIT Bombay
[Jun'23-Mar'24]

Part of a club promoting biotechnology in the college by conducting many events based on bioinformatics, forensics, etc.

- Designed a disease modelling segment for Bio-engineering GC & a focused genomics module for **70+ students**
- Organized successful BioX events including 'CSI' (with **70+ participants**) & 'Arcano' (with **40% higher turnout**)
- Delegated for BioX club at **Energy Swaraj Foundation's '1000 Days' event**, supporting vital sustainability outreach

EXTRACURRICULARS AND TECHNICAL SKILLS

Extra-Curricular Activities

- Trained in **Classical Vocals** under **NSO** for one year and performed at **Commencement'23** & **Republic Day'23**
- Created and presented a new watch-version of LinkedIn using Figma in **Vision'24**, the flagship event of Design Club
- Participated in **Crossy GC** & represented Hostel 6 by completing a challenging **5.5 km marathon** across the campus

Technical Skills

Programming & DS: NumPy, Pandas, SciPy, Scikit-Learn, Seaborn, Matplotlib, MySQL, MS Excel (Pivot Tables)
Softwares: SimuLink, LTSpice, Ngspice, AnyLogic, AutoCAD Fusion 360, ANSYS, LaserCAD, Fraktory, Arduino IDE